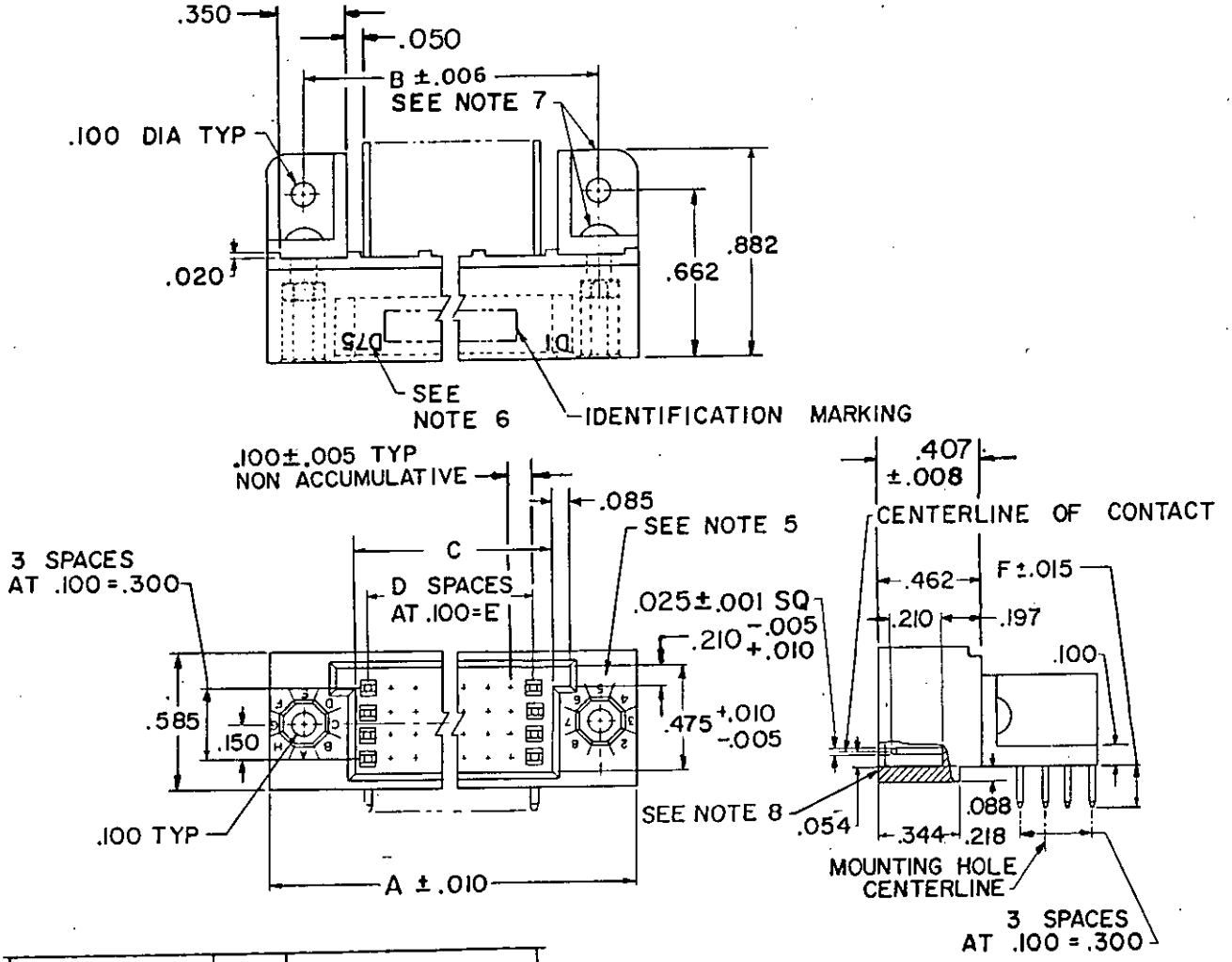
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES:
4-ROW PIN ASSEMBLIES, RIGHT ANGLE, 100 THROUGH 684
CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

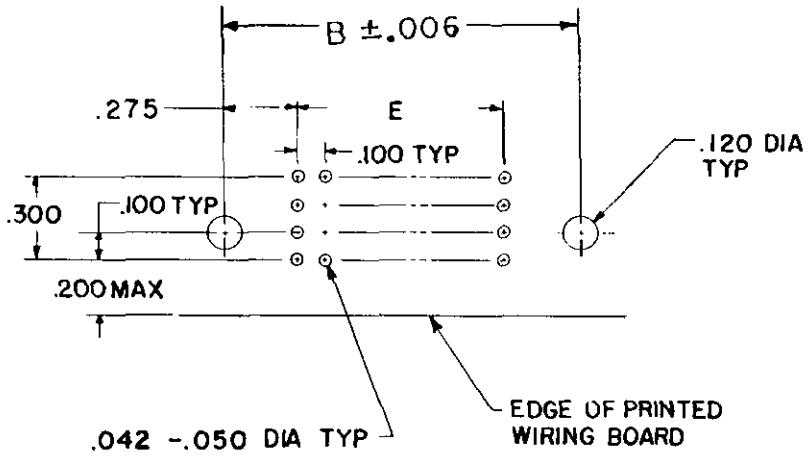


F ±.015		Type	Type of termination
Inches	mm		
.180	4.57	I	Solder post 1/
.165	4.19	II	Solder post
.140	3.56	III	Solder post
.120	3.05	IV	Solder post

1/ Solder post shall be .036 (0.91 mm) maximum diameter/or .036 (0.91 mm) maximum across the diagonal.

FIGURE 1. Connectors, receptacle assemblies .100 (2.54 mm) spacing, 100 through 300 contacts.

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DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.



Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.025	0.64
.037	0.94
.043	1.09
.050	1.27
.054	1.37
.088	2.24
.100	2.54
.120	3.05
.150	3.81
.182	4.62
.197	5.00
.200	5.08
.210	5.33
.218	5.54
.300	7.62
.344	8.74
.350	8.89
.407	10.34
.462	11.73
.475	12.06
.585	14.86
.662	16.81
.882	22.40

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/180.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row closest to polarizing feature, and markings every ten positions stamped this surface.
7. Mounting ears and hardware are purchased separately. See MIL-C-55302/182 for mounting hardware.
8. Chamfer shall be .015 minimum at a 45° angle ±.005 degrees.

FIGURE 1. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing, 100 through 300 contacts - Continued.

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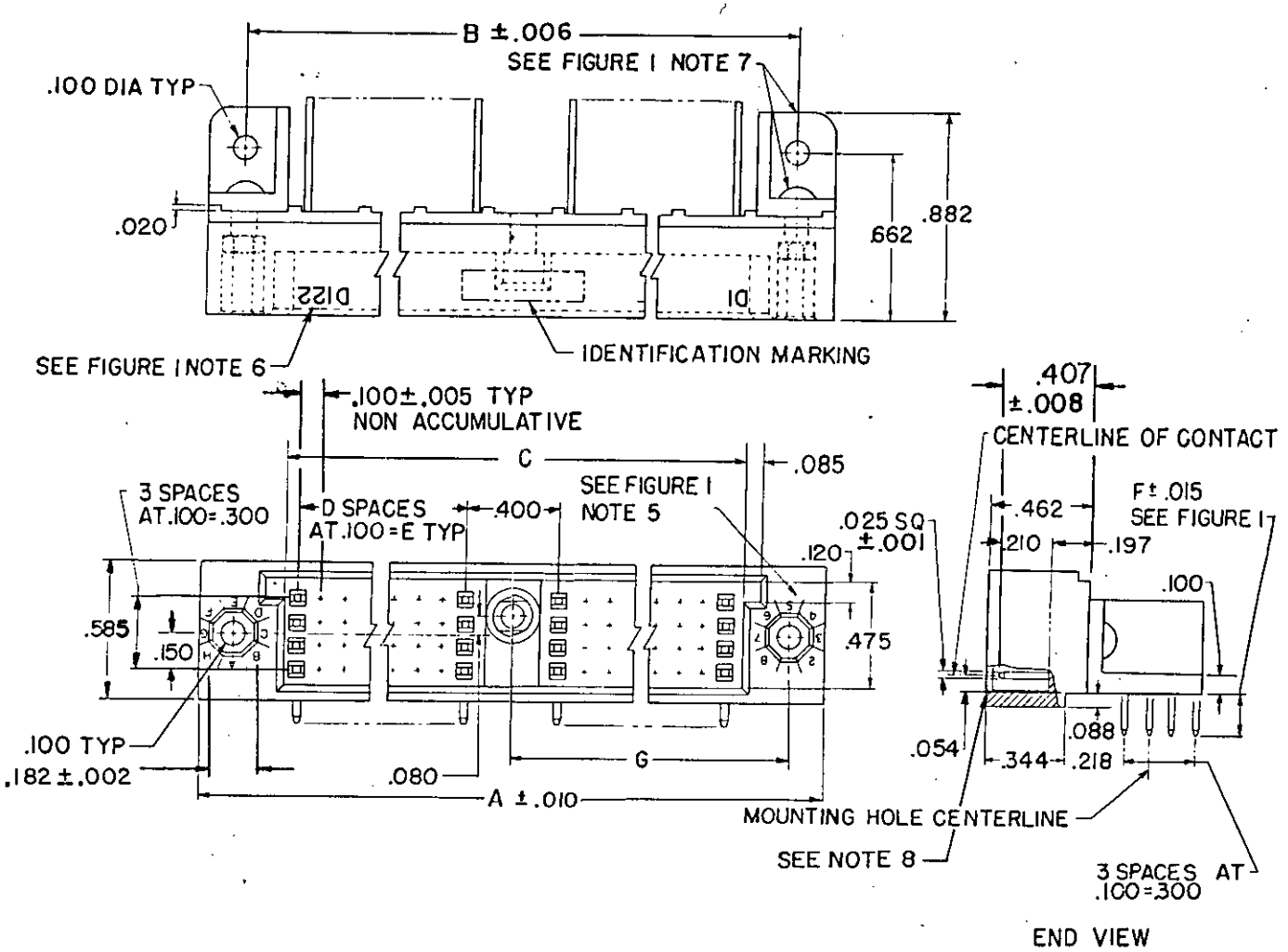
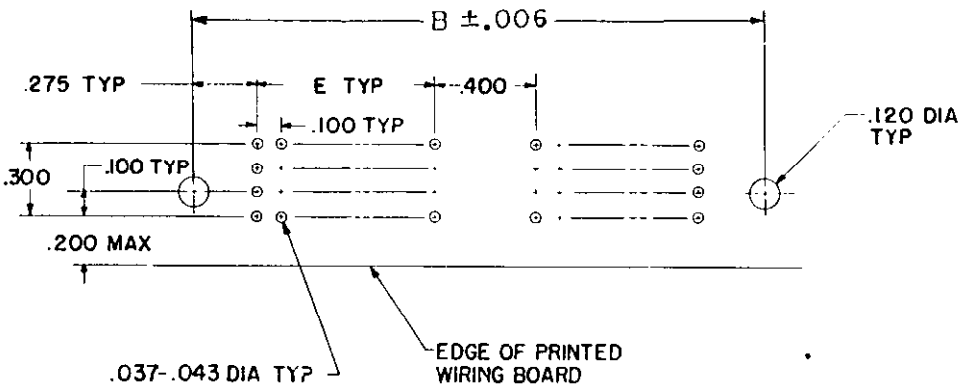


FIGURE 2. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing with one guide 320 through 488 contacts.

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Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.042	1.07
.050	1.27
.054	1.37
.080	2.03
.085	2.16
.088	2.24
.100	2.54
.120	3.05
.150	3.81
.182	4.62
.197	5.00
.210	5.33
.218	5.54
.275	6.98
.300	7.62
.344	8.74
.400	10.16
.407	10.34
.475	12.06
.585	14.86
.662	16.81
.882	22.40

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With two guides, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 2. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing with one guide, 320 through 488 contacts - Continued.

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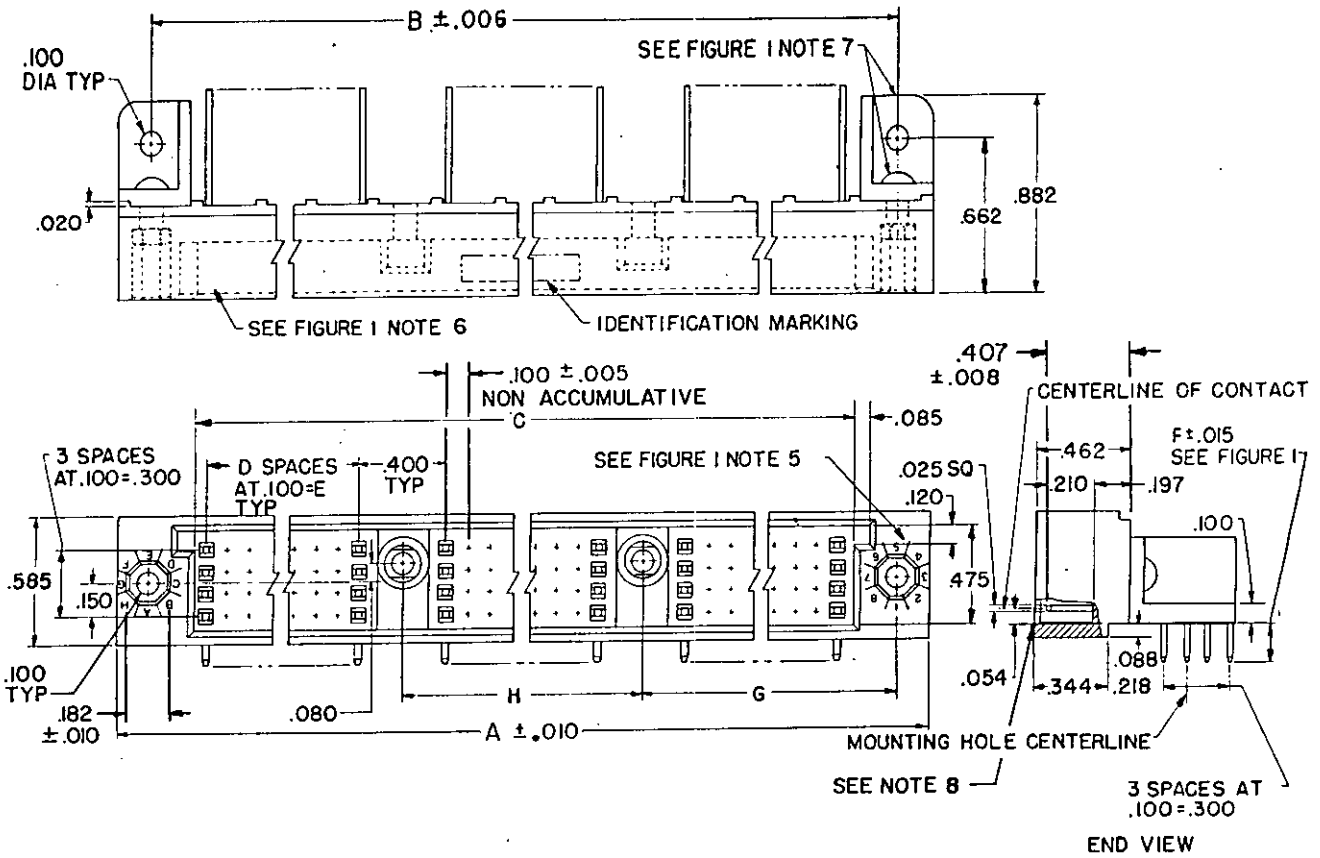
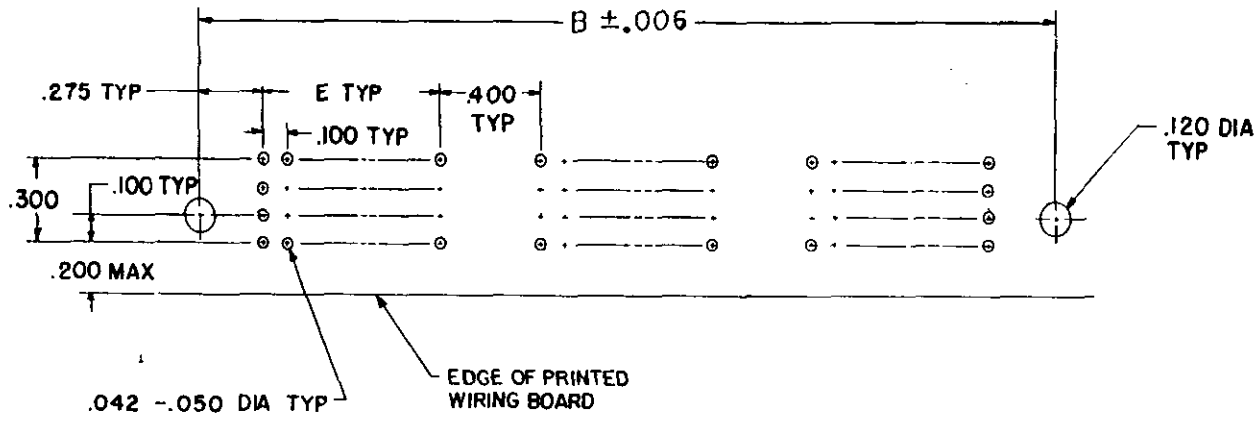


FIGURE 3. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing with two guides 516 through 684 contacts - Continued.

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Recommended circuit layout

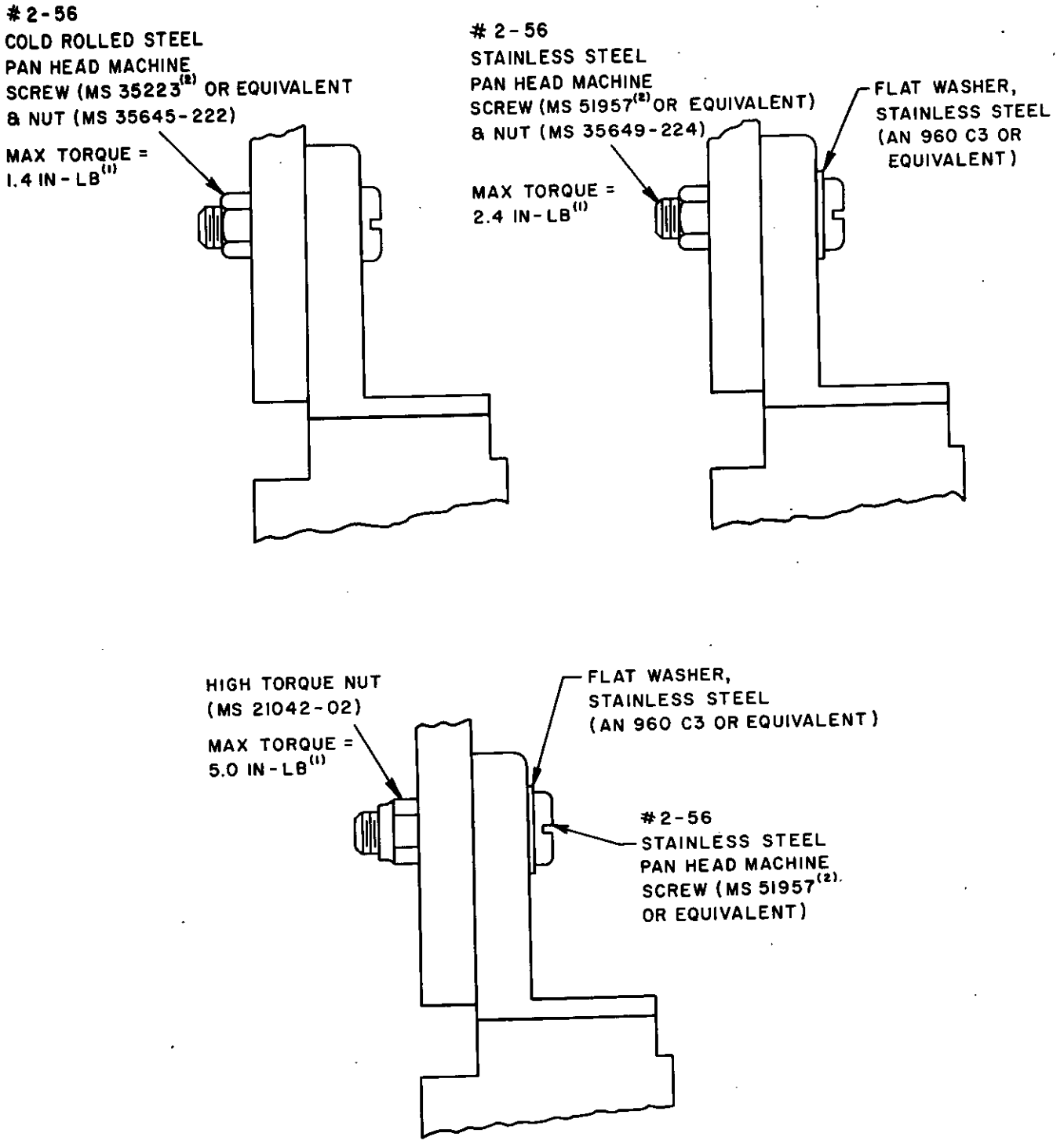
Inches	mm
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.020	0.51
.025	0.64
.042	1.07
.050	1.27
.054	1.37
.080	2.03
.085	2.16
.088	2.24
.100	2.54
.120	3.05
.150	3.81
.182	4.62
.197	5.00
.210	5.33
.218	5.54
.300	7.62
.344	8.74
.400	10.16
.407	10.34
.462	11.73
.475	12.06
.585	14.86
.662	16.81
.882	22.40

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With one guide, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 3. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing with two guides, 516 through 684 contacts - Continued.

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NOTES:
1. Torque wrench must be used when fastening connector to printed circuit board.
2. Length to be determined by user.

FIGURE 4. Mounting recommendations.

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TABLE I. Dash number, number of contacts, contact type, and dimensions.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 02 03 04	100	1	I II III IV	3.250 (82.55)	2.950 (74.93)	2.524 (64.11)	24	2.400 (60.96)			1-25
05 06 07 08	120	1	I II III IV	3.750 (95.25)	3.450 (87.63)	3.024 (76.81)	29	2.900 (73.66)			1-30
09 10 11 12	128	1	I II III IV	3.950 (100.33)	3.650 (92.71)	3.224 (81.89)	31	3.100 (78.74)			1-32
13 14 15 16	140	1	I II III IV	4.250 (107.95)	3.950 (100.33)	3.524 (89.51)	34	3.400 (86.36)			1-35
17 18 19 20	160	1	I II III IV	4.750 (120.65)	4.450 (113.03)	4.024 (102.21)	39	3.900 (99.06)			1-40
21 22 23 24	180	1	I II III IV	5.250 (133.35)	4.950 (125.73)	4.524 (114.91)	44	4.400 (111.76)			1-45
25 26 27 28	200	1	I II III IV	5.750 (146.05)	5.450 (138.43)	5.024 (127.61)	49	4.900 (124.46)			1-50
29 30 31 32	220	1	I II III IV	6.250 (158.75)	5.950 (151.13)	5.524 (140.31)	54	5.400 (137.16)			1-55
33 34 35 36	240	1	I II III IV	6.750 (171.45)	6.450 (163.83)	6.024 (153.01)	59	5.900 (149.86)			1-60
37 38 39 40	260	1	I II III IV	7.250 (184.15)	6.950 (176.53)	6.524 (165.71)	64	6.400 (162.56)			1-65

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TABLE I. Dash number, number of contacts, contact type, and dimensions.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
41 42 43 44	268	1	I II III IV	7.450 (189.23)	7.150 (181.61)	6.724 (170.79)	66	6.600 (167.64)			1-67
45 46 47 48	280	1	I II III IV	7.750 (196.85)	7.450 (189.23)	7.024 (178.41)	69	6.900 (175.26)			1-70
49 50 51 52	300	1	I II III IV	8.250 (209.55)	7.950 (201.93)	7.524 (191.11)	74	7.400 (187.96)			1-75
53 54 55 56	320	2	I II III IV	9.050 (229.87)	8.750 (222.25)	8.324 (211.43)	39	3.900 (99.06)	4.375 (111.13)		1-80
57 58 59 60	344	2	I II III IV	9.650 (245.11)	9.350 (237.49)	8.924 (226.67)	42	4.200 (106.68)	4.675 (118.75)		1-86
61 62 63 64	368	2	I II III IV	10.250 (260.35)	9.950 (252.73)	9.524 (241.91)	45	4.500 (114.30)	4.975 (126.37)		1-92
65 66 67 68	392	2	I II III IV	10.850 (275.59)	10.550 (267.97)	10.124 (257.15)	48	4.800 (121.92)	5.275 (133.99)		1-98
69 70 71 72	416	2	I II III IV	11.450 (290.83)	11.150 (283.21)	10.724 (272.39)	51	5.100 (129.54)	5.575 (141.61)		1-104
73 74 75 76	440	2	I II III IV	12.050 (306.07)	11.750 (298.45)	11.324 (287.63)	54	5.400 (137.16)	5.875 (149.23)		1-110
77 78 79 80	464	2	I II III IV	12.650 (321.31)	12.350 (313.69)	11.924 (302.87)	57	5.700 (144.78)	6.175 (156.85)		1-116

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TABLE I. Dash number, number of contacts, contact type, and dimensions.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
81 82 83 84	488	2	I II III IV	13.250 (336.55)	12.950 (328.93)	12.524 (318.11)	60	6.000 (152.40)	6.475 (164.46)		1-222
85 86 87 88	516	3	I II III IV	14.250 (361.95)	13.950 (354.33)	13.524 (343.51)	42	4.200 (106.68)	4.675 (118.75)	4.600 (116.84)	1-129
89 90 91 92	540	3	I II III IV	14.850 (377.19)	14.550 (369.57)	14.124 (358.75)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135
93 94 95 96	564	3	I II III IV	15.450 (392.43)	15.150 (384.81)	14.724 (373.99)	46	4.600 (116.84)	5.075 (128.91)	5.000 (127.00)	1-141
97 98 99 100	588	3	I II III IV	16.050 (407.67)	15.750 (400.05)	15.324 (389.23)	48	4.800 (121.92)	5.275 (133.99)	5.200 (132.08)	1-147
101 102 103 104	612	3	I II III IV	16.650 (422.91)	16.350 (415.29)	15.924 (404.47)	50	5.000 (127.00)	5.475 (139.07)	5.400 (137.16)	1-153
105 106 107 108	636	3	I II III IV	17.250 (438.15)	16.950 (430.53)	16.524 (419.71)	52	5.200 (132.08)	5.675 (144.15)	5.600 (142.24)	1-159
109 110 111 112	660	3	I II III IV	17.850 (453.39)	17.550 (445.77)	17.124 (434.95)	54	5.400 (137.16)	5.875 (149.23)	5.800 (147.32)	1-165
113 114 115 116	684	3	I II III IV	18.450 (468.63)	18.150 (461.01)	17.724 (450.19)	56	5.600 (142.24)	6.075 (154.31)	6.000 (152.40)	1-171

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TABLE II. Contact plating requirements.

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish: Contact engagement area shall have a minimum plating in accordance with MIL-C-55032. Solder post area (contact type I, .195 minimum length; type II, .180 minimum length; type III, .155 minimum length; type IV, .135 minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.
C	Overall contact finish: Shall be palladium or palladium nickel alloy (86/14 ±6 percent), 50 microinches minimum thickness, with a gold flash of 3 microinches minimum, over a nickel underplate of 50 microinches minimum in accordance with QQ-N-290, and shall meet the requirements of MIL-P-45209.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1, 2, 3, 4 and table I.

Material: In accordance with MIL-C-55302.

Plating: See table II.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31):

Two keys, part number MS55302/31-04, and two .086-56 UNC-2A mounting screws, are recommended. Keys and screws shall be ordered separately.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .094; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

Part number: M55302/181 (and dash number from table I and type from table II).

Example: M55302/181-01A.

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Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR
Navy - AS, OS, SH
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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